THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angles of a triangle are $(60 - d)^\circ$, 60° and $(60 + d)^\circ$. When $d = 25$, the largest angle is: 1

(a) 105° **(b)** 85° **(c)** 75° **(d)** 95°

2. In two triangles, $AB = PQ$, $BC = QR$, $CA = RP$. They are congruent by: 1

(a) ASA **(b)** AAA **(c)** SAS **(d)** SSS

3. An isosceles triangle has equal sides 15 cm and base 8 cm. Its perimeter (cm) is: 1

(a) 40 **(b)** 38 **(c)** 39 **(d)** 37

4. In a right-angled triangle one acute angle is 40°. The exterior angle at the other acute vertex is: 1

(a) 150° **(b)** 140° **(c)** 130° **(d)** 120°

5. The angles of a triangle are in the ratio $2 : 5 : 11$. The largest angle is: 1

(a) 110° **(b)** 130° **(c)** 120° **(d)** 100°

6. A base angle of an isosceles triangle is 70°. The exterior angle at the apex is: 1

(a) 160° **(b)** 140° **(c)** 150° **(d)** 130°

7. The exterior angle at B of $\triangle ABC$ is 150° and $\angle A = 85^\circ$. Then $\angle C$ is: 1

(a) 55° **(b)** 65° **(c)** 75° **(d)** 85°

8. Two sides of a triangle are 5 cm and 5 cm. The number of possible whole-number lengths of the third side is: 1

(a) 9 **(b)** 8 **(c)** 11 **(d)** 10

9. The angles of a triangle are $(60 - d)^\circ$, 60° and $(60 + d)^\circ$. When $d = 15$, the largest angle is: 1

(a) 75° **(b)** 85° **(c)** 95° **(d)** 65°

10. Two triangles have all three angles equal. They are: 1

(a) congruent by SSS **(b)** congruent by ASA
(c) not necessarily congruent **(d)** congruent by SAS

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In isosceles $\triangle ABC$, $AB = AC$ and the exterior angle at B is 120°. Find all three angles. 2

12. An isosceles triangle has perimeter 18 cm and one side 4 cm. Find its other sides (check which case is possible). 2

13. The exterior angle of a triangle is $(4x - 4)^\circ$ and the interior opposite angles are $(x + 16)^\circ$ and $(2x + 18)^\circ$. Find x and the exterior angle. 2

14. In isosceles $\triangle ABC$, $AB = AC$ and the exterior angle at B is 140°. Find all three angles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of $\triangle ABC$ are in the ratio $2 : 5 : 11$. Find the angles and the exterior angle at the vertex of the largest angle. 3

16. In $\triangle PQR$, $PQ = PR$, $\angle Q = (3x - 12)^\circ$ and $\angle R = (x + 8)^\circ$. Find x, all three angles and the exterior angle at P. 3

17. $\triangle ABC \cong \triangle DEF$ with $AB = (8x + 5)$ cm, $DE = (3x + 30)$ cm, $BC = 9$ cm and $AC = 12$ cm. Find x, the length DE and the perimeter of $\triangle DEF$. 3

OR

The angles of $\triangle ABC$ are in the ratio $1 : 1 : 2$. Find the angles and the exterior angle at the vertex of the smallest angle. 3

18. In $\triangle PQR$, $PQ = PR$, $\angle Q = (3x - 18)^\circ$ and $\angle R = (x + 10)^\circ$. Find x, all three angles and the exterior angle at P. 3

OR

$\triangle ABC \cong \triangle DEF$ with $AB = (9x + 3)$ cm, $DE = (4x + 33)$ cm, $BC = 10$ cm and $AC = 7$ cm. Find x, the length DE and the perimeter of $\triangle DEF$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The equal sides of an isosceles triangle are $(2x + 1)$ cm and $(3x - 9)$ cm and its base is $(x + 5)$ cm. Find x, the three sides and the perimeter. Check that the triangle exists. 5

OR

Four sticks of lengths 5 cm, 6 cm, 11 cm and 12 cm are available. Choosing any three, how many different triangles can be made? Test each choice. 5

20. The equal sides of an isosceles triangle are $(2x + 5)$ cm and $(3x - 3)$ cm and its base is $(x + 4)$ cm. Find x, the three sides and the perimeter. Check that the triangle exists. 5

OR

Four sticks of lengths 3 cm, 4 cm, 7 cm and 12 cm are available. Choosing any three, how many different triangles can be made? Test each choice. 5

END OF QUESTION PAPER — Total: 40 Marks